

APPLICATION

1. Cancun's daily high temperature, in degrees Celsius, can be modelled by the function

$$t(d) = -7 \cos \frac{2\pi}{365} d + 22, \text{ where } d \text{ is day of the year and } 1 = \text{January } 1 \text{ and } t(d) \text{ is}$$

temperature. On days when the temperature is 24° or above, the air conditioners at the resorts are turned on. How many days are the air conditioners running at the resorts over a span of 500 days beginning January 1? (round to the nearest day, a graph has been provided for you)

$$24 = -7 \cos \frac{2\pi}{365} d + 22$$

$$\cos \frac{2\pi}{365} d = \frac{2}{7}$$

$$RAA = 1.28$$

$$\frac{2\pi}{365} d = 1.86, 4.42, 8.14$$

$$d = 108.08, 256.92, 473.08$$

$$ANS = (x_2 - x_1) + (500 - x_3)$$

$$= 256.92 - 108.08 + 500 - 473.08$$

$$= 148.84 + 26.92 = 175.76 \approx 176 \text{ about } 176 \text{ days}$$

2. Refer back to the previous question; Round to 2 decimal places and make sure to include units

- a) Determine the average rate of change between day 200 and 220.

$$\frac{t(220) - t(200)}{220 - 200} = \frac{27.5914556102 - 28.684767284}{20}$$

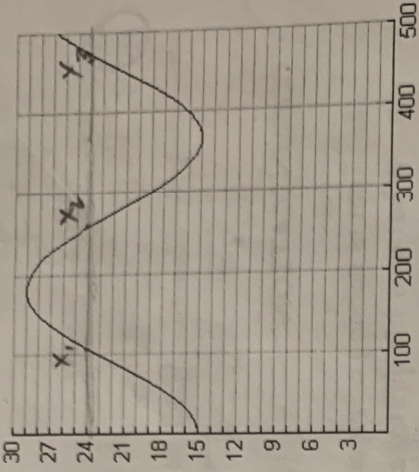
$$= -0.05 \text{ } ^\circ\text{C/day}$$

- b) Determine the instantaneous rate of change at day 220.

$$\frac{t(220.001) - t(220)}{220.001 - 220} =$$

$$\frac{27.59138314 - 27.59145561}{0.001}$$

$$= -0.07 \text{ } ^\circ\text{C/day}$$



$\therefore$ the air conditioners have been running for about 176 days

KNOWLEDGE

Total

10/10

1. a) Sketch the graph of $g(x) = -4 \cos\left[\frac{2}{3}x + \frac{\pi}{12}\right] + 1$ for one cycle on the grid provided.

(6)

- b) Determine:

period: $\frac{2\pi}{\frac{2}{3}} = 3\pi$ (0.5)

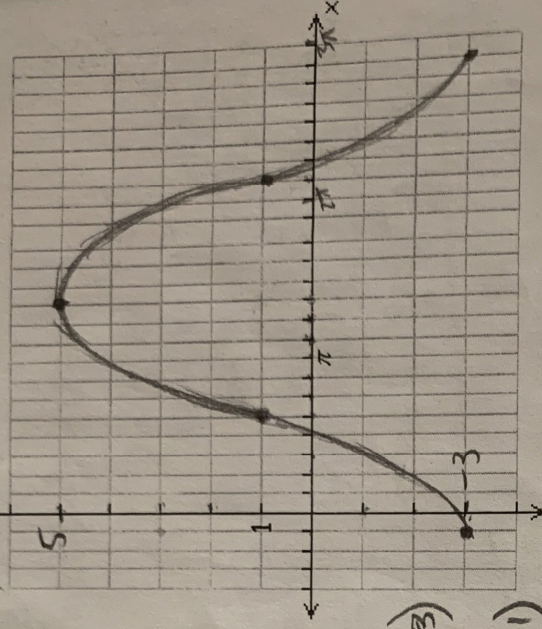
phase shift: $\frac{1}{8}\pi$ left (0.5)

vertical translation: 1 up (0.5)

range: $[-3, 5]$ (0.5)

y-int: $-\sqrt{6} - \sqrt{2} + 1 = -2.864$ (1)

mapping: $(x, y) \rightarrow \left(\frac{3}{2}x - \frac{1}{8}\pi, -4y + 1\right)$ (1)



$$(0, 1) \rightarrow \left(-\frac{\pi}{8}, 5\right)$$

$$\left(\frac{\pi}{2}, 0\right) \rightarrow \left(\frac{5\pi}{8}, 1\right)$$

$$(\pi, -1) \rightarrow \left(\frac{11\pi}{8}, 5\right)$$

$$\left(\frac{3\pi}{2}, 0\right) \rightarrow \left(\frac{17\pi}{8}, 1\right)$$

$$(2\pi, 1) \rightarrow \left(\frac{23\pi}{8}, -3\right)$$

$$= -4 \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) + 1$$

$$= -4 \left(\cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4}\right) + 1$$

$$= -4 \left(\frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}\right) + 1$$

$$= -4 \left(\frac{\sqrt{2} + \sqrt{6}}{2\sqrt{2}}\right) + 1$$

$$= -\sqrt{2} - \sqrt{6} + 1$$

2. Use the given graph to determine the following:

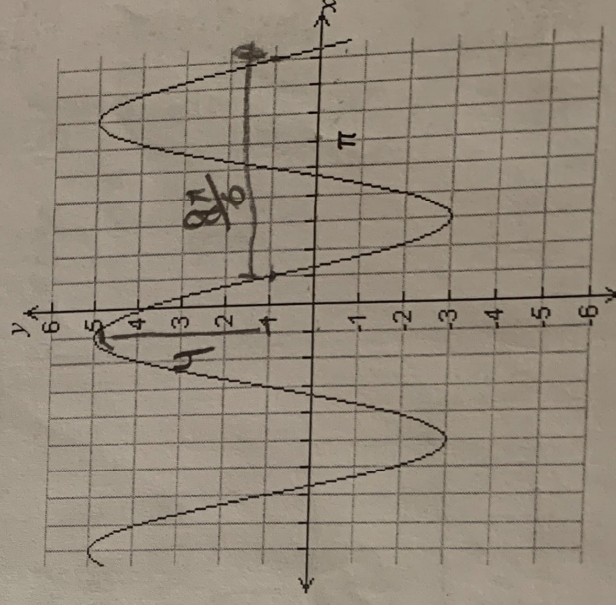
$$y = 1$$

$$a = 4$$

$$p = \frac{8}{6}\pi = \frac{4\pi}{3}$$

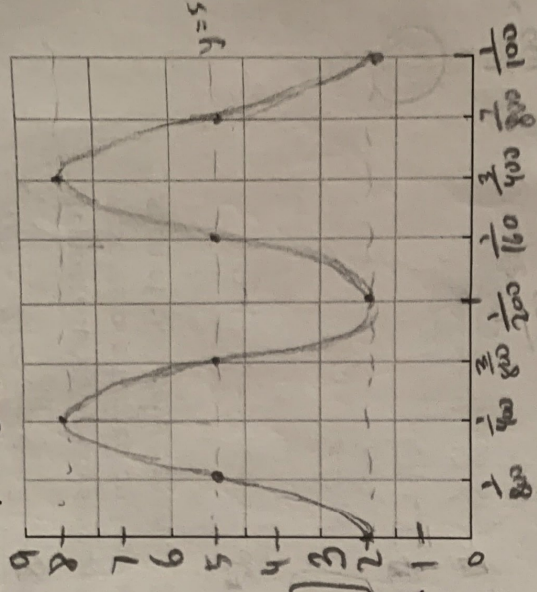
Sine equation: $y = 4 \sin\left(\frac{3}{2}\left(x - \frac{\pi}{6}\right)\right) + 1$

Cosine equation: $y = 4 \cos\left(\frac{3}{2}\left(x + \frac{\pi}{6}\right)\right) + 1$



3. The height of the tip of an airplane propeller above the ground once the airplane is at full speed can be modeled by a sinusoidal function. At full speed, the 3 meter propeller makes 200 revolutions per second. At $t=0$, the tip of the propeller is at its minimum height which is 2m above the ground.

a) Sketch two cycles of a graph that represents the height of the tip of the propeller above the ground, as a function of time, in seconds. Label your graph.



b) Write an equation that describes this situation.

$$y = 2 + 3 = 5$$

$$\alpha = \frac{200 \text{ rev}}{s} = \frac{1 \text{ rev}}{\frac{1}{200} s}$$

height (m)

$$P = \frac{1}{200} \text{ sec}$$

$$K = \frac{2\pi}{\frac{1}{200}} = 400\pi$$

$$y = 3 \cos(400\pi x) + 5$$

your graph is correct

COMMUNICATION

1. Tom is solving a trigonometric equation where $0 \leq x \leq 2\pi$.

What was his mistake? Explain why his reasoning is incorrect, then give the correct solution.

Tom divided by $\cos x$, imposing an additional restriction of $\cos x \neq 0$, but in the case of this question $\cos x = 0$ is a valid solution, which made him miss some solutions.

$$3 \cos^2 x - 2 \cos x = 0$$

$$\cos x (3 \cos x - 2) = 0$$

case ① $\cos x = 0$

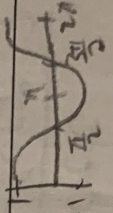
$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$3 \cos^2 x = 2 \cos x$$

$$3 \cos x = 2$$

$$\cos x = \frac{2}{3}$$

$$x = 0.84 \quad x = 5.44$$



case ② $\cos x = \frac{2}{3}$
 $\text{RAA} = 0.84$
 $x = 0.84, 5.44$

2. State two points where the function $y = -2 \sin(2\pi x) + 7$ has an instantaneous rate of change that is zero.

The rate of change will be 0 at either the maximum or minimum points, because the tangent line is horizontal at those points.

Max is 9

$$9 = -2 \sin(2\pi x) + 7$$

$$\sin(2\pi x) = -1$$

$$2\pi x = \frac{3\pi}{2}$$

$$x = \frac{3}{4}$$

min is at 5

$$5 = -2 \sin(2\pi x) + 7$$

$$-2 = -2 \sin(2\pi x)$$

$$\sin(2\pi x) = 1$$

$$2\pi x = \frac{\pi}{2}$$

$$x = \frac{1}{4}$$

$\therefore$ 2 points where the instantaneous rate of change is 0 is at $x = \frac{1}{4}$ and $x = \frac{3}{4}$

EXPONENTIAL THINKING

- Solve for x on the interval $0 \leq x \leq 2\pi$. Give the exact solution(s) in radian measure, where possible. Otherwise, round to the nearest hundredth of a radian.

b) $1 = -\sin x \cot x + 3 \sin^2 x$

$$1 = -\sin x \cot x + 3 \sin^2 x$$

because $\sin x \neq 0$
 otherwise division by 0

$$= -\cos x + 3(1 - \cos^2 x)$$

$$= -\cos x + 3 - 3 \cos^2 x$$

$$3 \cos^2 x + \cos x - 2 = 0$$

$$(3 \cos x - 2)(\cos x + 1) = 0$$

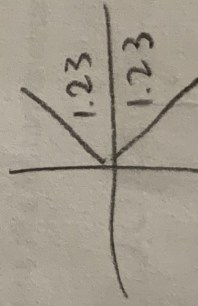
$$b) \left[3 \cos^2 \left(x - \frac{\pi}{3} \right) - 3 \sin^2 \left(x - \frac{\pi}{3} \right) \right] = 1$$

$$3 \left[\cos^2 \left(x - \frac{\pi}{3} \right) - \sin^2 \left(x - \frac{\pi}{3} \right) \right] = 1$$

$$3 \cos \left(2 \left(x - \frac{\pi}{3} \right) \right) = 1$$

$$\cos \left(2 \left(x - \frac{\pi}{3} \right) \right) = \frac{1}{3}$$

$$RAA = 1.23$$



$$2 \left(x - \frac{\pi}{3} \right) = -1.23, 1.23, 5.05, 7.51$$

$$x - \frac{\pi}{3} = -0.615, 0.615, 2.525, 3.757$$

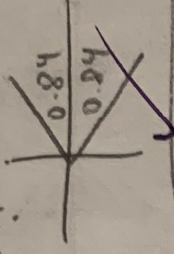
$$x = 0.43, 1.66, 3.57, 4.80$$

case ①

$$3 \cos x + 2 = 0$$

$$\cos x = -\frac{2}{3}$$

$$RAA = 0.84$$



$$x = 0.84, 5.44$$

case ②

$$\cos x = -1$$

$$\cos x = 1$$

$$x = \pi$$

but since $\sin \pi = 0$

this solution is impossible

as $\sin x \neq 0$

$$\therefore x = 0.84 \text{ or } x = 5.44$$

$$0 \leq x \leq 2\pi$$

$$-\frac{\pi}{3} \leq x - \frac{\pi}{3} \leq \frac{5\pi}{3}$$

$$-\frac{2\pi}{3} \leq 2 \left(x - \frac{\pi}{3} \right) \leq \frac{10\pi}{3}$$

$$-2.09 \leq 2 \left(x - \frac{\pi}{3} \right) \leq 10.47$$

only need to check solutions of $\cos$ in this